

NORTHEASTERN, NC, USA

WMO: 723074

Lat: 36.028N

Lon: 76.567W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03703

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	26.9	3.0	6.4	31.4	8.6	8.6	33.1	23.3	48.3	20.2	46.7	4.9	0	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	93.3	77.9	91.2	77.5	89.7	76.9	81.1	89.2	79.8	87.3	78.6	85.6	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.0	85.3	77.6	143.4	83.6	76.7	139.2	82.9	44.5	88.8	43.1	87.0	42.2	85.8	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.0	16.8	14.8	16.8	96.8	3.4	2.8	14.3	98.8	12.3	100.4	10.4	102.0	7.9	104.0		
			14.8	82.8	3.3	2.3	12.5	84.5	10.6	85.8	8.7	87.1	6.3	88.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	43.1	45.4	51.5	61.1	69.6	77.2	80.5	79.4	73.4	63.6	53.2	46.6
	DBStd	15.40	10.29	9.28	9.36	8.45	7.33	5.63	4.67	4.50	5.74	8.15	8.41	8.76
	HDD50	790	266	184	97	8	0	0	0	0	0	4	58	173
	HDD65	2943	680	550	427	173	42	2	0	0	7	126	363	573
	CDD50	5225	53	56	145	342	606	817	945	912	701	426	155	67
	CDD65	1902	2	3	10	56	183	369	480	447	258	83	9	2
	CDH74	16891	7	16	67	395	1348	3367	4985	4326	1857	486	34	3
	CDH80	5749	0	0	5	80	380	1194	1952	1562	485	89	2	0
Wind	WSAvg	6.4	6.7	7.1	7.6	8.0	7.0	6.5	6.1	5.3	5.6	5.6	5.7	5.9
Precipitation	PrecAvg	48.4	3.9	3.4	4.1	3.2	4.1	4.5	5.7	5.3	4.6	3.5	2.7	2.9
	PrecMax	67.3	9.2	7.9	8.7	7.0	9.8	10.9	13.7	11.8	18.9	10.9	5.7	6.8
	PrecMin	32.7	0.4	1.1	1.0	0.6	0.9	0.9	1.9	0.9	0.2	0.0	0.8	0.4
	PrecStd	7.7	1.9	1.7	1.8	1.5	2.2	2.1	2.9	2.8	3.6	2.8	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.9	76.8	79.8	86.4	91.2	95.1	96.9	95.3	90.6	86.3	78.6	73.1
		MCWB	66.7	65.4	63.4	70.3	75.4	77.3	79.6	79.3	75.4	74.6	67.1	64.8
	2%	DB	68.0	71.0	75.2	81.9	87.6	91.5	93.7	92.7	88.0	82.2	73.2	69.5
		MCWB	61.2	60.8	61.9	66.9	73.1	76.0	79.2	78.4	75.2	71.3	64.6	64.1
	5%	DB	63.6	65.6	71.8	79.0	84.1	90.0	91.2	90.4	85.0	79.3	70.9	64.6
		MCWB	58.2	58.4	60.8	66.0	71.8	75.7	78.3	78.1	74.9	69.8	63.8	60.9
	10%	DB	60.1	61.4	66.9	74.9	81.4	87.1	89.6	87.9	82.3	76.2	66.9	61.8
		MCWB	55.3	54.7	59.3	64.5	70.2	74.7	77.7	76.7	73.6	68.9	61.6	56.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.1	67.3	68.1	73.5	77.3	80.3	83.7	82.5	79.8	77.3	70.4	67.5
		MCDB	71.4	73.7	74.6	80.8	86.0	88.4	93.8	91.8	85.6	81.4	74.7	70.6
	2%	WB	63.3	63.5	65.5	70.9	75.4	78.8	81.5	80.8	78.4	75.2	67.8	64.7
		MCDB	66.2	67.6	71.8	77.3	83.4	87.2	89.8	89.0	83.7	79.6	71.9	67.3
	5%	WB	59.6	59.9	63.0	68.6	73.9	77.6	80.2	79.4	77.2	73.0	65.1	61.9
		MCDB	63.0	64.8	68.3	75.5	81.4	85.6	87.9	86.8	82.4	77.2	69.0	64.8
	10%	WB	55.7	55.5	60.4	66.4	72.5	76.5	78.9	78.3	75.8	70.4	62.4	58.1
		MCDB	59.0	60.5	66.1	73.1	79.3	83.9	86.1	85.0	80.9	74.7	65.7	61.3
Mean Daily Temperature Range	5% DB	MDBR	17.0	18.8	19.5	19.6	17.4	16.0	14.9	14.8	15.4	18.2	19.1	17.7
		MCDBR	21.4	23.9	23.4	22.0	19.0	17.5	16.2	16.1	16.5	18.7	20.4	21.0
		MCWBR	16.7	16.0	12.9	10.7	8.2	7.0	7.0	6.8	7.2	9.5	13.3	16.9
	5% WB	MCDBR	20.0	22.4	19.7	18.9	17.1	15.4	14.7	14.7	13.0	14.4	17.5	19.0
		MCWBR	17.2	17.3	14.1	11.5	8.2	6.9	7.3	6.8	6.4	8.4	13.7	17.1
Clear-Sky Solar Irradiance	taub	0.308	0.324	0.350	0.391	0.438	0.493	0.521	0.509	0.413	0.360	0.327	0.315	
	taud	2.539	2.490	2.444	2.337	2.238	2.131	2.067	2.091	2.360	2.494	2.535	2.552	
	Ebn at Noon	284	291	292	283	270	254	246	246	267	274	274	274	
	Edh at Noon	25	30	34	39	44	49	52	50	36	29	25	23	
All-Sky Solar Radiation	RadAvg	791	1024	1345	1715	1903	2007	1904	1723	1417	1161	882	703	
	RadStd	57	111	99	124	145	108	100	101	102	136	65	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+293	+198

Nomenclature: See separate page